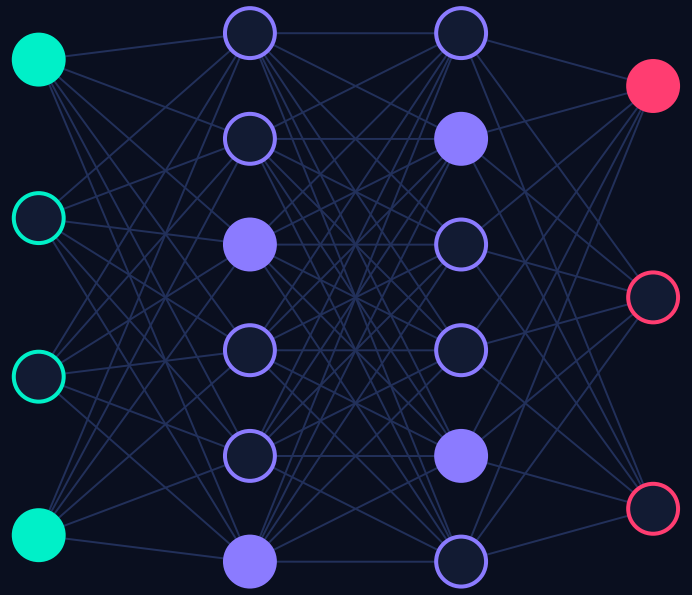
AI + CYBERSECURITY | OCT 2026

AEGISNET

A self-explaining Edge-AI neural network that catches cyber-attacks, even ones it has never seen.



Built from scratch in NumPy | 34 KB | 12/12 tests passed | 100% free tools

Presented by: Rushi | Analyst

Oct 2026: models race, production lags

40%

of enterprise apps will embed task-specific AI agents by end-2026 (Gartner)

11%

of firms run AI agents in production, although 79% say they have adopted them

~9 in 10

organisations now use AI regularly in at least one function (McKinsey 2026)

3x

more usage of small task-specific models than giant LLMs by 2027 (Gartner)

THE MODEL RACE (early Oct 2026)

Claude Opus 5.5 | GPT-6.1 | Gemini 4 | DeepSeek V4.1 | Qwen3.8 | Kimi K3

Violet = closed frontier Aqua = open-weight

Frontier models are converging. The winners are small, explainable, governed systems that actually ship, which is exactly what AegisNet is.

The gap: what IT teams really need



Small and private

Runs on a laptop or Raspberry Pi. Telemetry never leaves your network.



Explainable

On-call engineers need to know WHY they were paged, not just a score.

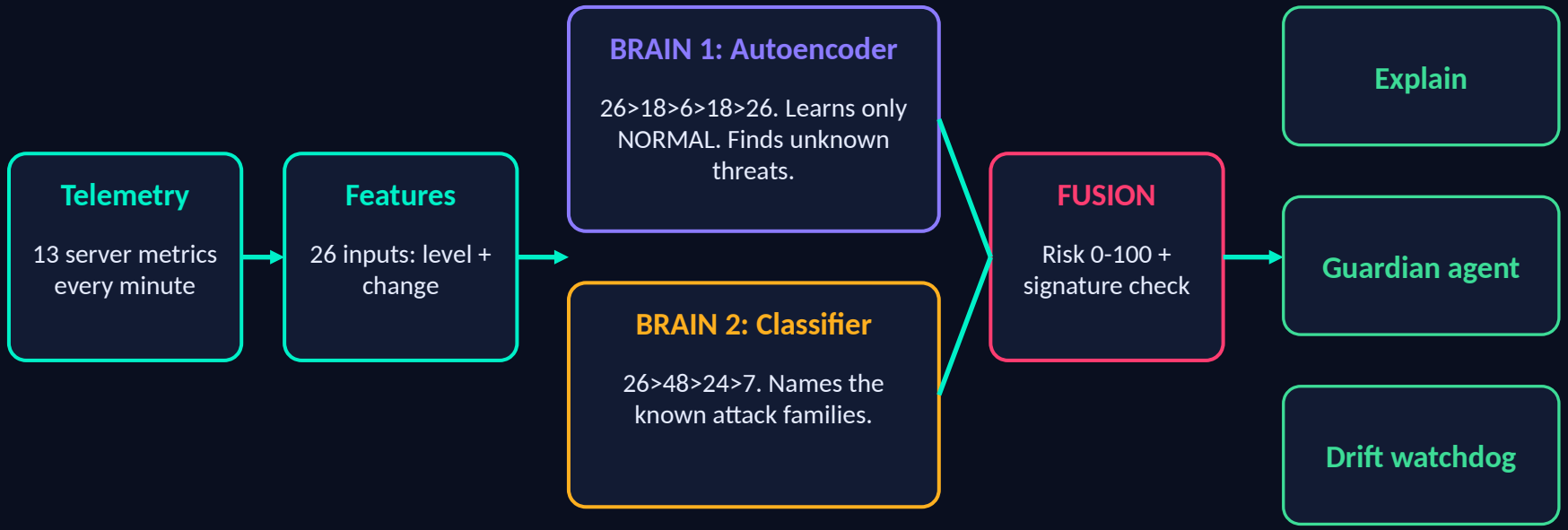


Governed

Risky actions need a human OK, and the model must warn when it goes stale.

Our answer: one 34 KB neural network that watches server metrics, explains every alert, and asks a human before anything risky.

Architecture: the Dual-Brain design



Brain 1 never sees an attack, so it flags anything strange (zero-days). Brain 2 recognises known attacks by name. Fusion adds a check that turns never-seen patterns into 'unknown anomaly'.

Six ideas that make it different

Dual-Brain

Unsupervised and supervised networks fused into one 0-100 risk score.

Zero-day safety net

Never-seen patterns become 'unknown anomaly', never a wrong label.

Explains itself

Every alert names the metrics that caused it.

Guardian agent

Safe actions auto-run. Risky ones wait for a human.

Drift watchdog

A PSI monitor says when the model needs retraining.

Edge-ready

34 KB, 0.004 ms per window. Fits a Raspberry Pi.

A realistic test bench

28,800 minutes of normal data to train Brain 1

28,800 minutes of labelled attacks for Brain 2

14,400 unseen test minutes, 37 attack episodes

- Daily load cycle, correlated metrics
- Benign flash-crowds as hard negatives
- Attack strength 40-100%: weak ones hide

Attack	Tell-tale metrics
Brute-force	failed logins, connections up
DDoS	inbound traffic, connections, HTTP 5xx up
Port scan	unique ports way up, tiny packets
Exfiltration	outbound traffic + DNS queries up
Cryptominer	CPU and process count up
Ransomware	file writes + disk I/O way up

Results on 10 unseen days

98.5%

attack-minute recall

0.98%

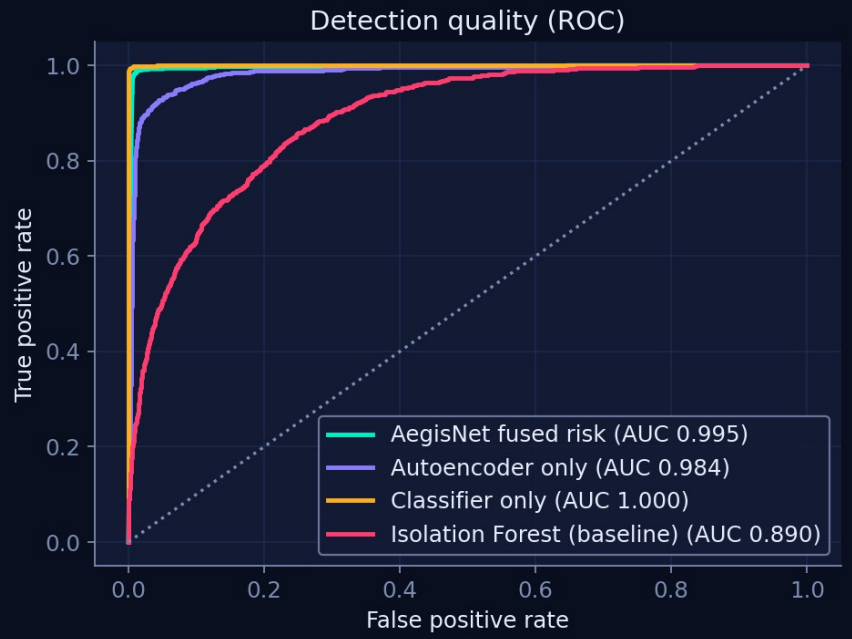
false-alarm rate

37/37

attack episodes detected

0.2 min

mean time to detect



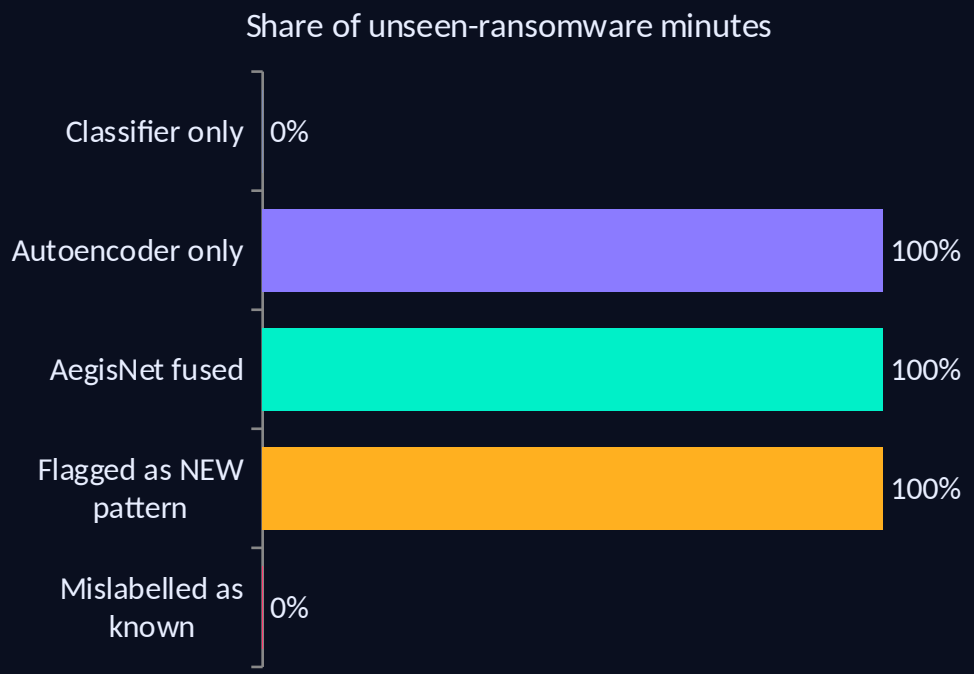
Fused AUC 0.995 vs Isolation-Forest baseline 0.890. Precision 83%: attacks are rare, so some alerts are false.

One day of live monitoring



Red = real attacks. Amber dots = AegisNet alerts. Over 10 days it raised 40 incidents; only 3 had no real attack behind them.

Zero-day test: the unseen ransomware



Setup

Ransomware was removed from the classifier's training data.

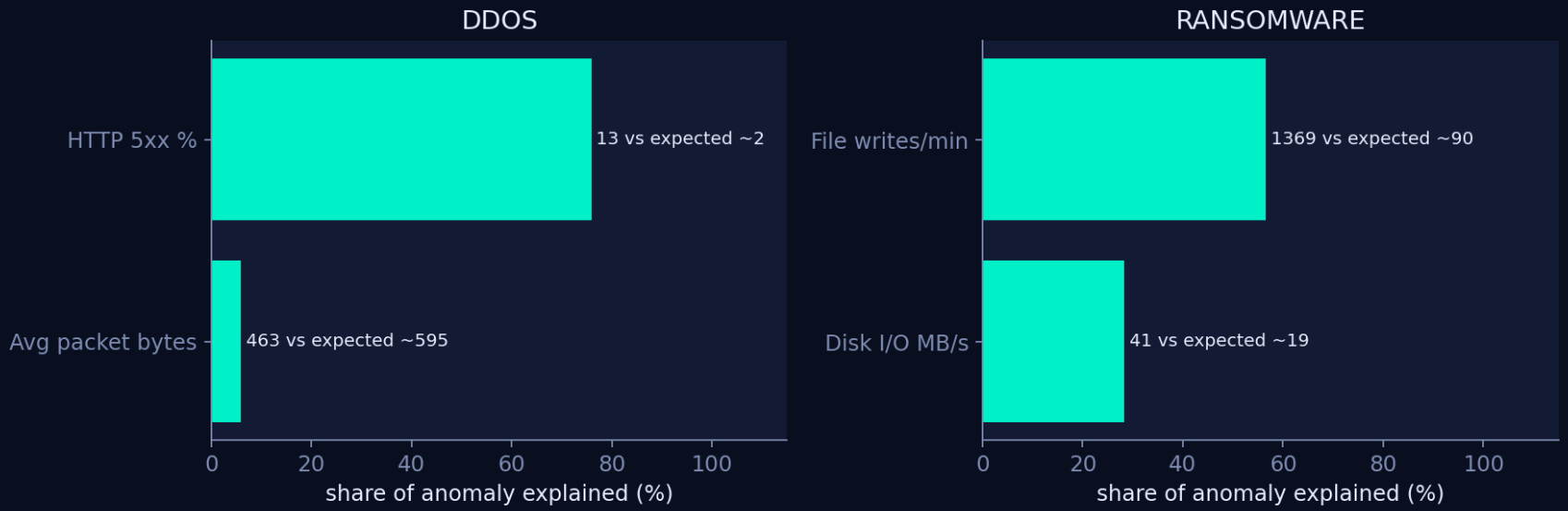
Result

The classifier alone was blind (0%). AegisNet caught every minute and tagged it a new pattern instead of guessing a wrong name.

Why

Brain 1 only knows what normal looks like.

Every alert explains itself



Bars show how much of the anomaly each metric explains. The label compares the observed value with what the autoencoder expected, so an engineer sees the cause in seconds.

Guardian agent: bounded autonomy

```
[CRITICAL] Volumetric DDoS attack
minutes 92666-92694 (29 min) | peak risk 100/100

WHY   HTTP 5xx % = 13 (model expected ~2)
      Avg packet bytes = 463 (model expected ~595)

AUTO  Enable CDN / WAF rate-limiting profile
AUTO  Page the on-call network engineer

ASK   Null-route top offending prefixes
ASK   Scale out replicas behind the load balancer

AUDIT 4 actions logged, 2 wait for a human
```



1 Detect

Risk crosses the alert line.



2 Auto-run safe actions

Snapshots, tickets, paging.

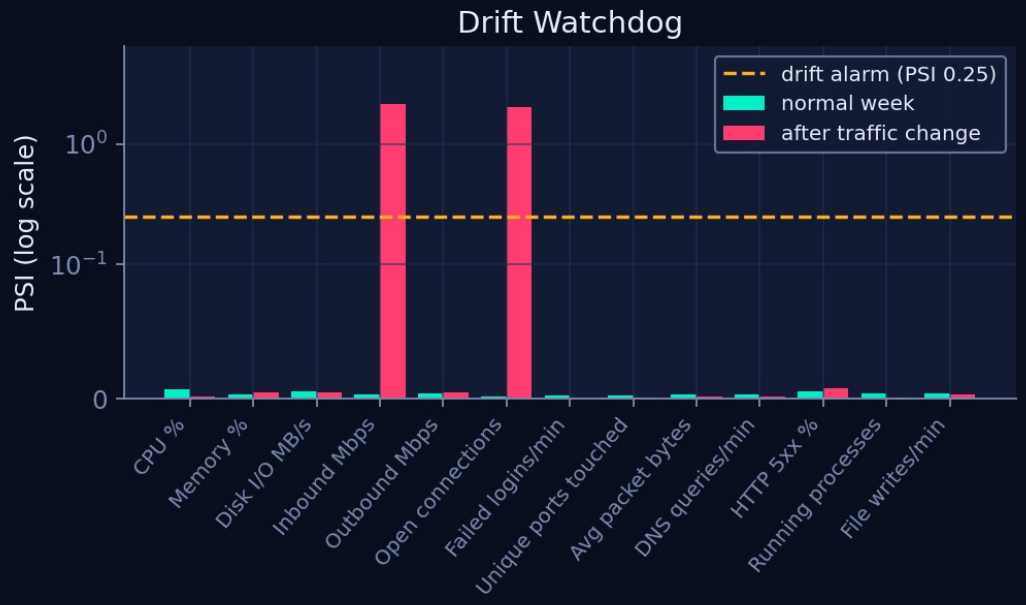


3 Human OKs risky ones

Block, isolate, quarantine.

Every action is audit-logged. Actions are simulated; connect your firewall or ticket API to go live.

Built for production, not just demos



Normal week: PSI 0.007 = STABLE. After a traffic change: PSI 2.15 on inbound traffic = retrain alarm.

34 KB

whole model file

0.004 ms

per minute-window on 1 CPU

12 / 12

automated tests pass + REST API

100% free tools, run it in 3 commands

Python 3 | NumPy | pandas | scikit-learn | Matplotlib | Flask | unittest

```
# 1. install
pip install -r requirements.txt

# 2. train + evaluate + report
python run_all.py

# 3. prove it works
python -m unittest discover -s tests -v
```

Tested, not claimed

- Gradient check proves the backprop maths
- Quality, zero-day, drift and API tests
- Verified from a clean folder, plus a live API call

What we proved, and what is next

Proved

- 98.5% attack recall at 1.0% false alarms
- Catches an attack family it never saw
- Explains and asks before acting
- Fits in 34 KB, runs for free

Honest limits and next steps

- Data is simulated: test on CICIDS2017 and real logs
- Guardian actions are simulated: plug in a firewall API
- Add a local LLM to write incident stories
- Export to ONNX, train federated across servers

Thank you. Questions?

Rushi | Analyst